



Toegepaste Micro-economie: Public economics

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Week 3: Asymmetric Information

3. Asymmetric information

First two weeks:

- General framework
- Market failure in case of public goods & externalities
 - Cost and benefits to people not involved in market transactions are ignored by consumers and producers
 - Market prices do not reflect these cost and benefits

3. Asymmetric information

Today: Market failure due to asymmetric information

Roadmap

- Incomplete vs. asymmetric information
- Market failure due to:
 - Hidden actions → moral hazard
 - Hidden characteristics → adverse selection
- Application to insurance markets

Asymmetric information

Distinction between:

- Imperfect information
 - Uncertainty faced by all parties in a transaction
 - Example: future stock price company
- Asymmetric information (or private information)
 - (At least) one party knows more about
 1. Its own **actions** affecting the value of the transaction
 - ‘Hidden actions’ leading to ‘Moral hazard’
 2. A **characteristic** affecting the value of the transaction
 - ‘Hidden characteristics’ leading to ‘Adverse selection’

Asymmetric information

Two main issues:

- Hidden actions → moral hazard
 - Non-verifiable actions that harm others in transaction
 - Employee slacking off; unnecessary repairs by car mechanic / dentist; Unhealthy living by individuals with health insurance
- Hidden characteristics → adverse selection
 - One party has better info on given aspect of transaction (that affects the value of the transaction)
 - Quality of second-hand items; probability of repaying loans;
Health insurance: Probability of needing treatment

Moral hazard

Undertaking non-verifiable, socially inefficient actions

- Actions beneficial to person choosing them, but cost to others **in transaction** are higher than this benefit.

→ Individually rational but socially inefficient

Leads to reduced efficiency of transaction, and may even completely obstruct transaction

Difference between moral hazard and external effects?

Moral hazard: example

Rent-a-scooter

If value of unsafe driving = 110, then scooter is rented, but outcome still inefficient

	Safe driving	Unsafe driving
Exp. cost to firm	50	100
Value to consumer	70	90

If this value > 120, then outcome is efficient

Efficient: rent scooter and drive safely

Outcome: no rental at all

- Firm anticipates that consumer will drive unsafe
→ Demands price ≥ 100
- Consumer refuses to rent the scooter at that price

Moral hazard

Moral hazard can lead to market failure:

As actions are not observable, market price does not differ between the case where an inefficient action was chosen versus the case where it was not.

→ Market price does not reflect all cost and benefits.

Moral hazard

Note 1:

If private gain $>$ cost to others, there is no problem

→ Action is efficient, and price adjusts to cover cost

If private gain $<$ cost to others, efficiency suffers

→ All parties involved could benefit from a commitment not to engage in socially inefficient actions

But: non-verifiability makes such commitment impossible

Note 2:

Moral hazard implies hurting others for personal gain. In reality, not everyone always engages in moral hazard

Moral hazard

Some research research:

- Dentists: Gottschalk et al (2020, Economic Journal)
- Taxi drivers: Balafoutas et al. (2013, Review of Economic Studies)

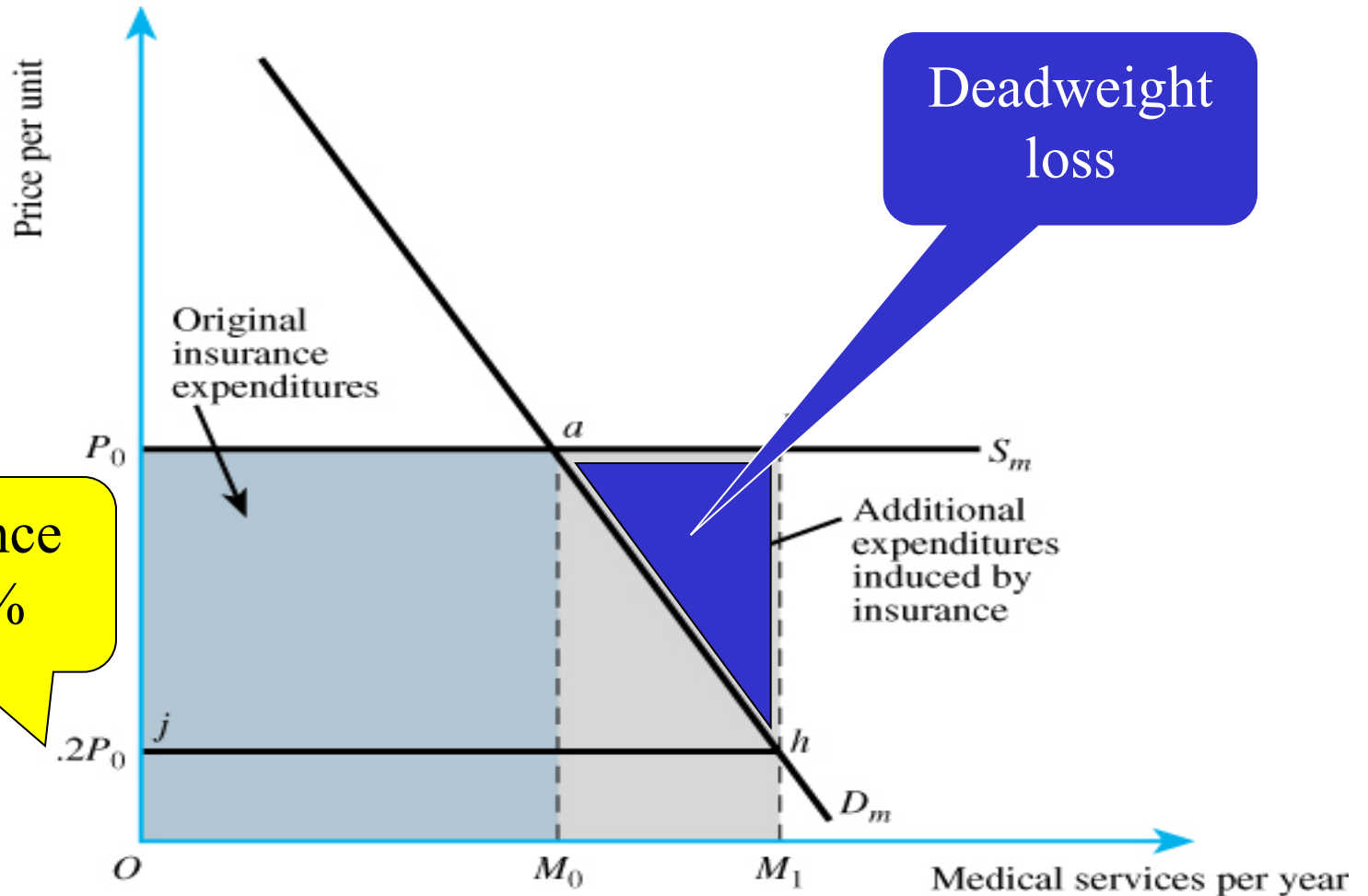
Moral hazard

Another example of moral hazard: increasing
‘consumption’ when other party pays (partially)

- More visits to physician by insured person
- More usage of public transport by students after introduction OV-card
- Students taking ‘too’ long to graduate, as society pays large part of cost
- Too much risk-taking by banks as they anticipate bail-outs when large losses arise

Note: externality arguments used to defend some of these policies → Trade-offs!

Figure 9.4



Moral hazard

Reality check (see book): Does demand for health care depend on insurance coverage?

RAND health insurance experiment ('74 – '81)

5800 US citizens allocated to 6 insurance plans

From free care up to 95% coinsurance rate (with cap)

Main finding (Aron-Dine, Einav, Finkelstein 2013):

People in more generous plans consume more care

In 95% coinsurance plan, average spending 39% lower than in free care plan

Moral hazard

How to improve efficiency?

Recall from last week: **Coase Theorem**

- If
- (Transferable) property rights assigned
 - Transaction cost sufficiently low

Then private parties will negotiate towards socially efficient outcome

However, this does not work in case of moral hazard: Actions are not (or hardly) verifiable
(agreement too costly to enforce → transaction cost!)
→ Contracts cannot provide commitment

Moral hazard

How can private parties improve efficiency?

- Monitoring
 - (Random) checks of activities; inspection. But: costly
- Paying for performance
 - no-claim bonus; requiring down-payments; piece-rates
 - But: may lead to inefficient risk-sharing or to distortions, if no perfect link between actions and measured performance
- Restricting set of possible actions (regulation)
 - Limits to insurance coverage; rigid working hours
 - But: Also restricts value of transaction
- Implicit contracts / reputation
 - ‘If you engage in moral hazard, I will never interact with you again’
 - Requires that actions are observable and that future suff. important
- Professional ethics / social norms:
 - ‘If I engage in moral hazard, I feel bad about myself.’

Moral hazard

Government intervention?

Moral hazard versus adverse selection

Recap:

Moral hazard reduces efficiency because individuals undertake **actions** that are individually rational but socially inefficient.

So, moral hazard is about (non-verifiable) actions

In contrast, **Adverse selection** can arise when people have ‘private’ information on relevant given **characteristics** (of people or objects)

So, adverse selection is about (fixed) characteristics (of people or objects) relevant to the transaction

Adverse selection

‘Hidden characteristics’:

Some (potential) parties in a transaction have better information about characteristics that affect the value of the transaction than others

Consequence: uninformed people cannot offer different prices for ‘good types’ and ‘bad types’ of trading partners with info

This can lead to adverse selection:

The informed parties self-select themselves into and out of transactions in such a way that the uninformed side gets the least valuable trading partners

Result:

Profitable transactions are not undertaken → inefficient.

Adverse selection: example

Rent-a-scooter

Two types of consumers: safe and unsafe.

Consumer knows its type, firm only knows $p(\text{safe}) = \frac{1}{2}$

	Safe type	Unsafe type
Exp. cost to firm	50	100
Value to consumer	70	120

Efficient: both types rent scooter

Outcome: Only unsafe types rent scooter: adverse selection

→ If both type would rent, expected cost 75

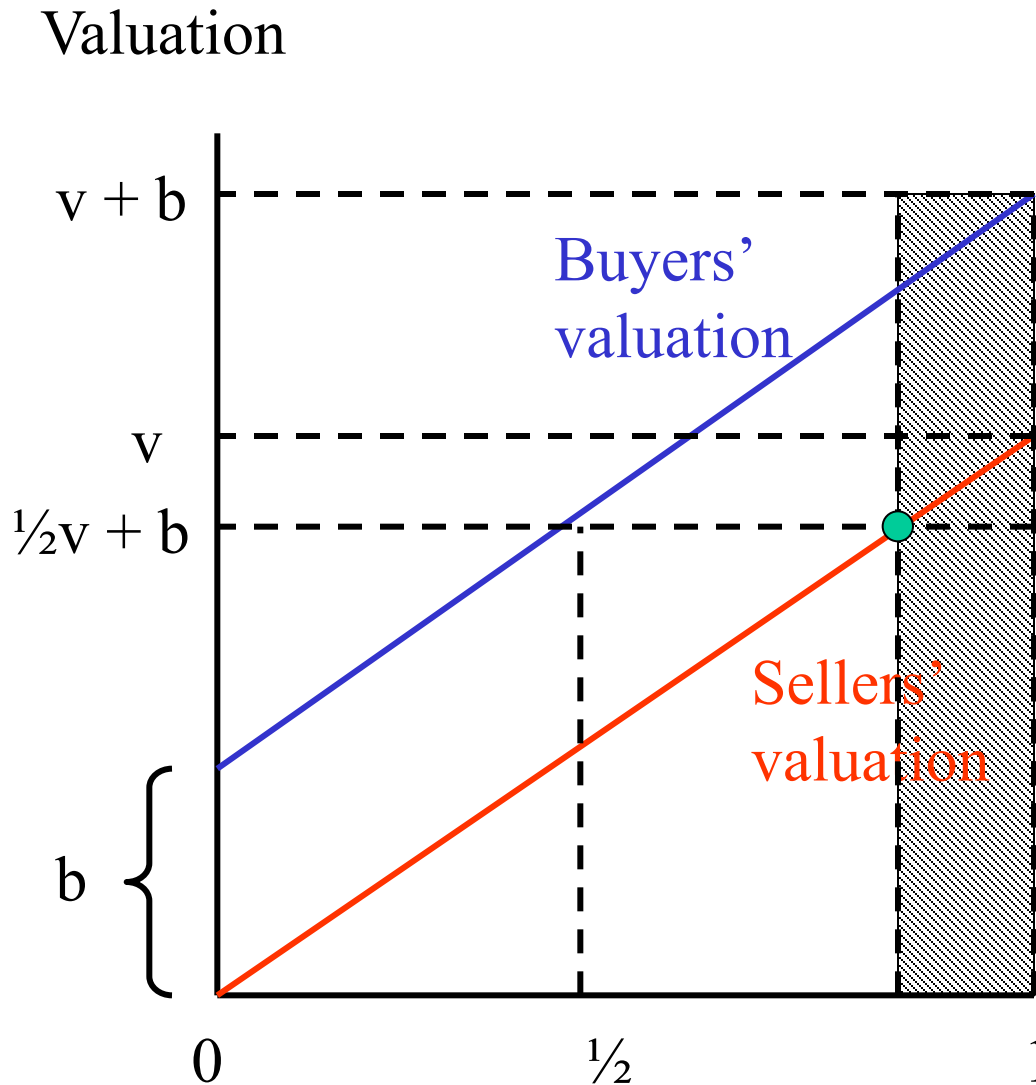
→ But safe type would not rent the scooter at that price

Adverse selection (video)

Market for second-hand cars:

- Second-hand cars differ in quality q
 - Uniformly distributed between 0 and 1.
 - There are as many buyers as sellers; all risk-neutral.
 - Sellers know the quality of their own car.
 - A seller's valuation of a car with quality q is vq
 - Buyers do not know the quality of a given car.
 - A buyers' valuation of a car with quality q is $vq + b$, where b is such that $0 \leq b < \frac{1}{2}v$
- Efficiency requires: all cars should be traded

Adverse selection (video)

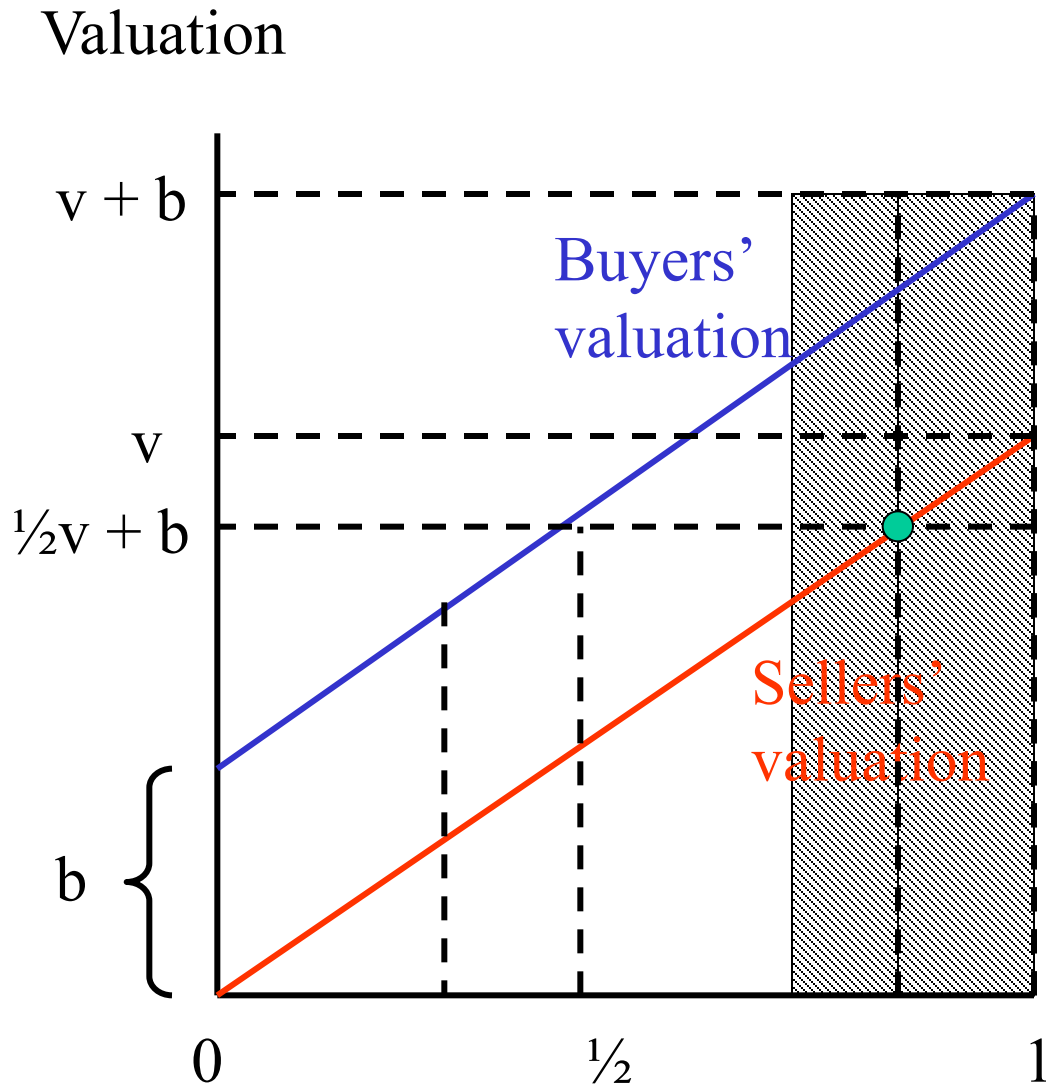


Crucial: price the same for all cars (independent of q), as buyers do not know q

Uniform distribution of q
→ average quality is $\frac{1}{2}$
→ if all cars traded, maximum price buyers are willing to pay is $\frac{1}{2}v + b$

But then, some sellers with high-quality cars refuse to sell

Adverse selection



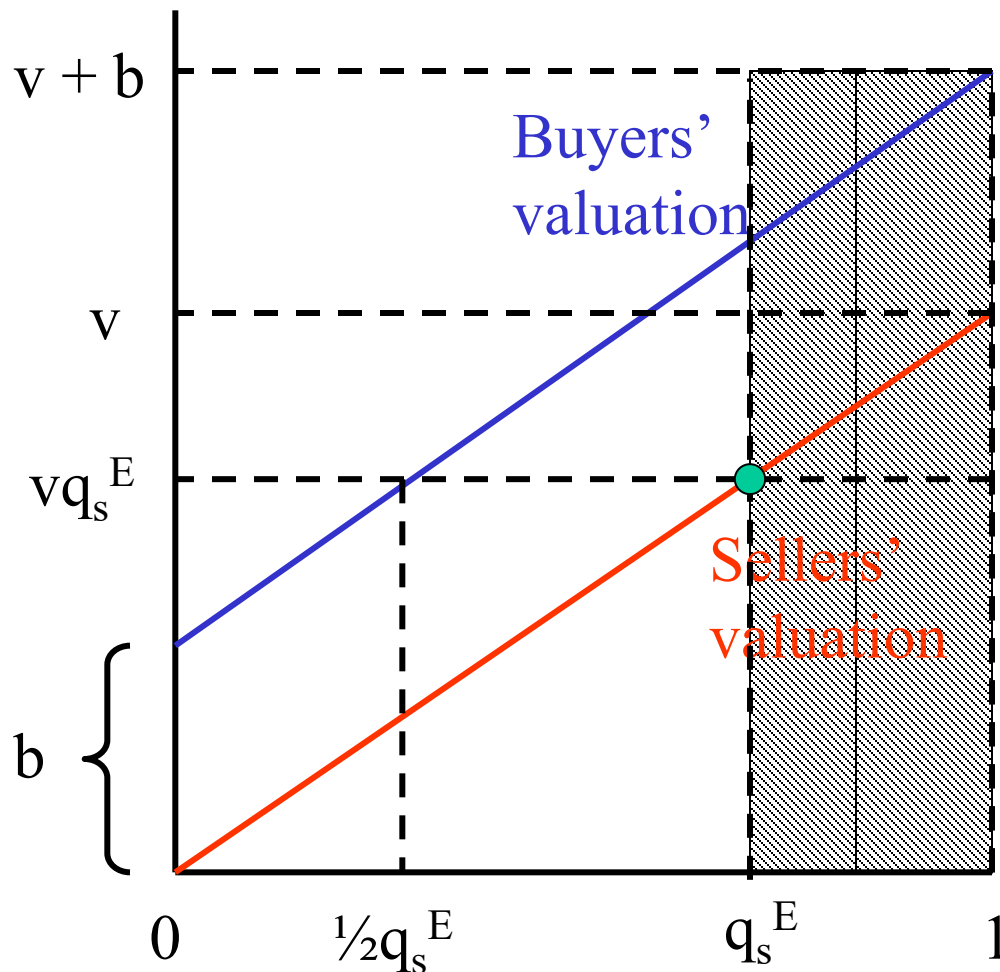
Anticipating this, buyers should have lower willingness to pay

Lower willingness to pay of buyers implies that even more sellers leave the market. etc.

Consequence:
Fraction of cars not traded $\rightarrow$ inefficient

Adverse selection (video)

Valuation



Equilibrium outcome?

Strategic interaction:
→ Nash- equilibrium!

How to find equilibrium?

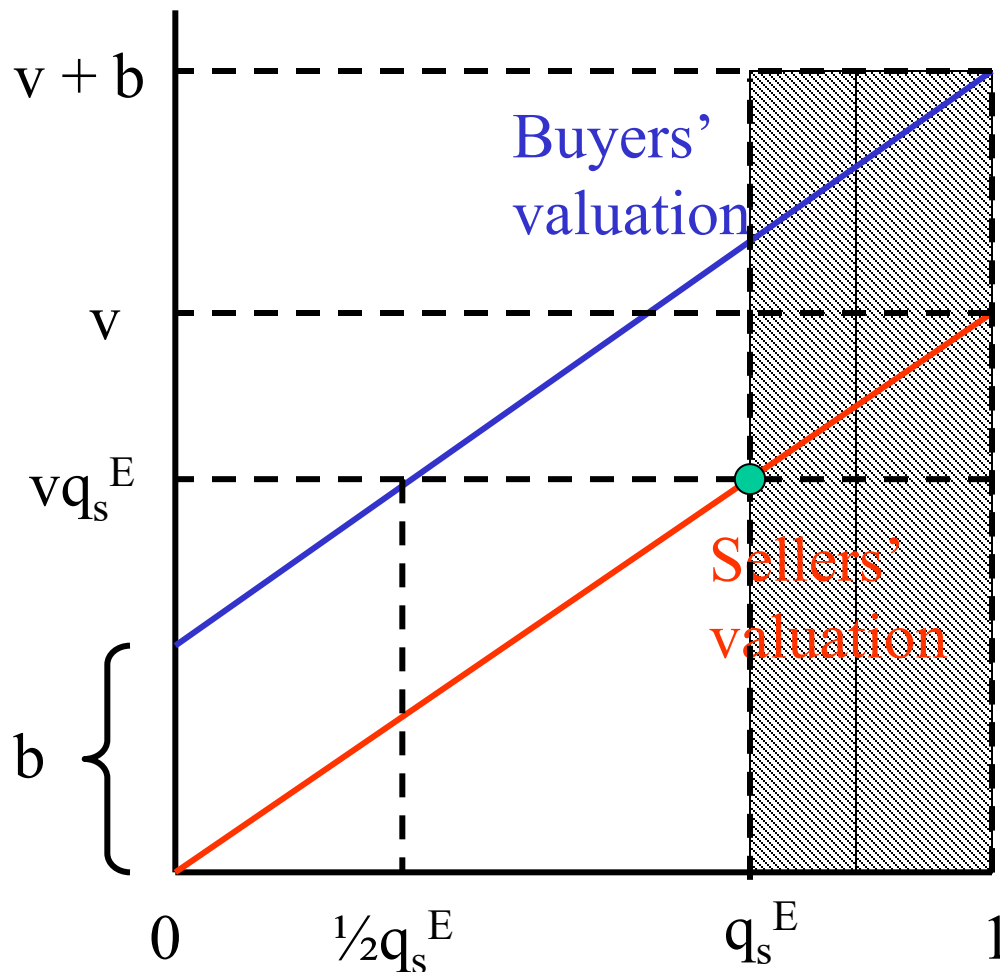
Crucial insight: given price in equilibrium, sellers with one particular q indifferent between selling and not selling!

Denote this by q_s^E

Car quality q

Adverse selection (video)

Valuation



Equilibrium outcome?

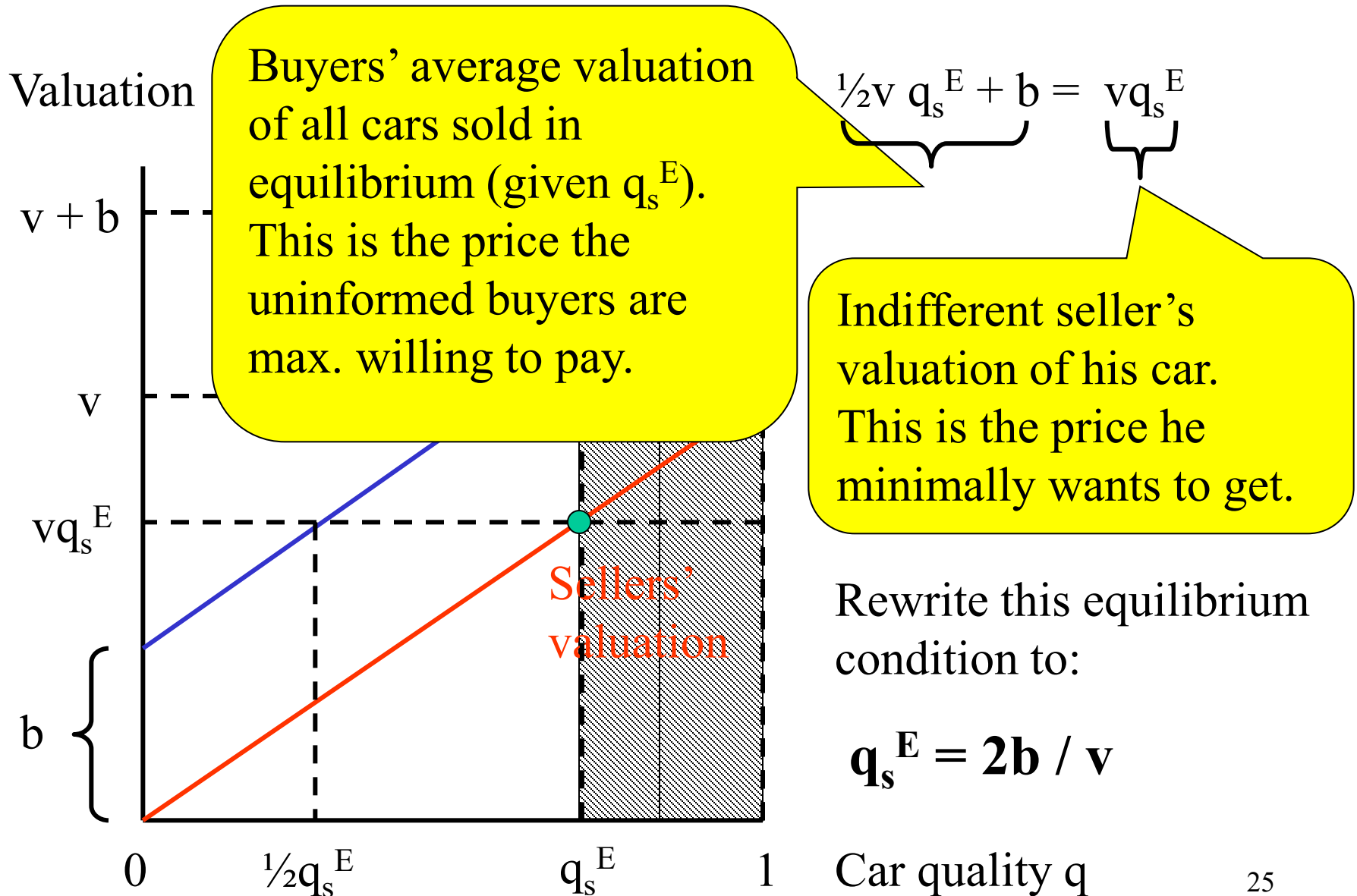
Given that all cars with $q \leq q_s^E$ are traded in equilibrium, buyers' average valuation of cars traded equals $\frac{1}{2} v q_s^E + b$

Hence, buyers willing to pay maximally $\frac{1}{2} v q_s^E + b$

In equilibrium, at that price, seller q_s^E must be indifferent

Car quality q

Adverse selection (video)



Adverse selection (video)

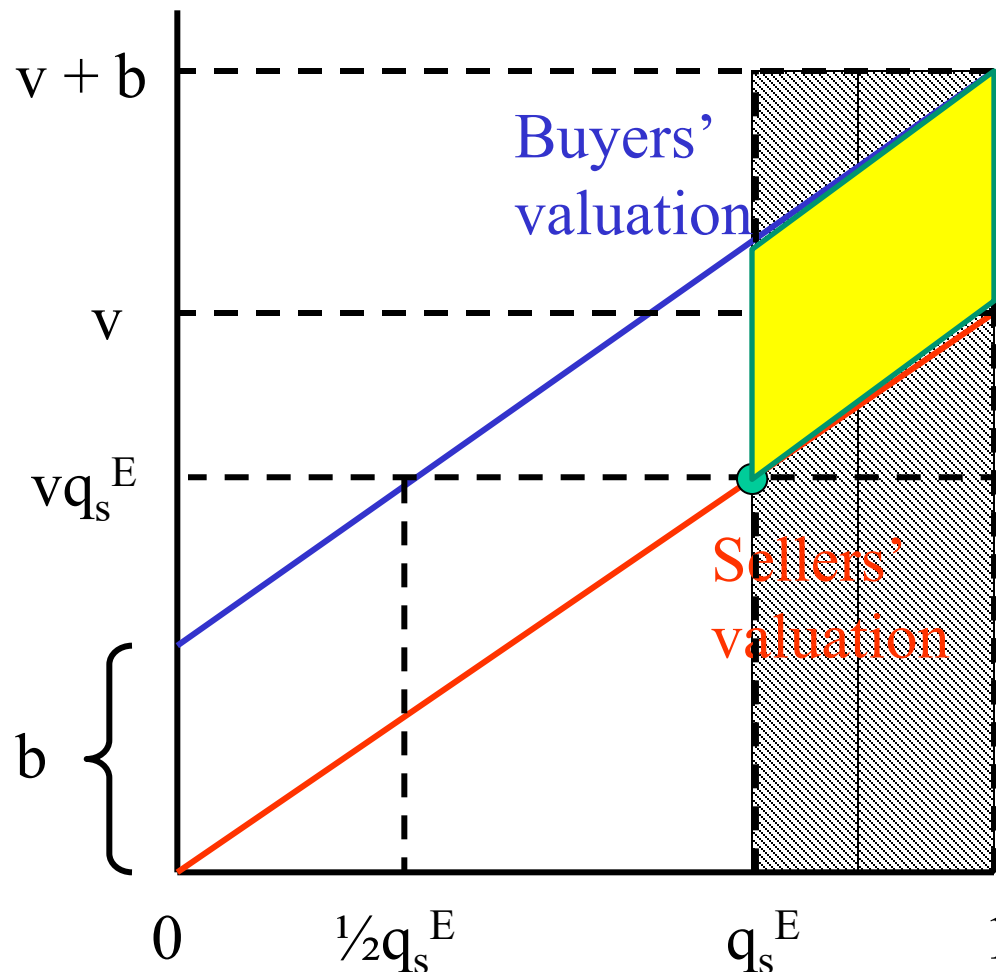
Hence, in equilibrium, only cars with $q \leq 2b/v$ are traded

Notes:

- In equilibrium, only part of the cars are traded.
→ Inefficient!
- More cars are traded if the difference between the valuation of buyers and sellers is large (b is high)
- The (uninformed!) buyers can only trade with sellers with relatively bad cars
→ adverse selection!

Adverse selection (video)

Valuation



Loss in efficiency =
Total value of all
transactions that do not
take place.

Here: b per transaction
(difference between
buyer's and seller's
valuation of a car)

So: efficiency loss =
 $b \cdot (1 - q_s^E) \cdot \text{total_}_\#__\text{of_cars}$

Adverse selection

Conclusion:

Hidden characteristics harm efficiency, as

1. Price / contract cannot depend on 'type' of informed party
2. Informed party self-selects into transaction such that types that yield low value to uninformed party are more likely to join
→ adverse selection
3. Which makes uninformed party hesitant to engage in transaction

Note: key assumption is that informed people are not honestly revealing their type

Empirical question: are people willing to lie for material gain?

Adverse selection

Possible solutions by private parties

- Uninformed parties have an incentive to search for info
 - Tests; expert opinions; peer reviews.
 - But: costly
- By offering multiple ‘contracts’, the uninformed party may induce the informed parties to self-select (‘screening’)
 - Additional insurance; offering fixed and variable wage
- High-quality informed parties want to ‘signal’ their quality
 - Warrantees; diploma’s and certificates
 - But again: costly
- Implicit contracts / establishing a reputation for honesty
 - ‘If you ever lie to me, I will never interact with you again’
 - Requires that information gets revealed (later) and that the future is sufficiently important

Asymmetric info and government intervention

Can the government reduce inefficiencies?

Note:

Inefficiencies arise from private parties lacking information

→ Government faces identical lack of information!

So, no easy solution through government intervention

Some measures that may reduce inefficiency:

- Make some actions illegal, with penalties (fraud; speeding)
- Make participation mandatory (car & health insurance)
- (Help to) provide info (quality standard; inspection)
- Public production ('poverty insurance' → next week)

Application: insurance

In many insurance markets, both moral hazard and adverse selection play a role

First: why insurance? Recall from Frank, ch. 6:

→ Risk-averse people dislike uncertainty in their income / wealth

Implication:

People with uncertain income are willing to pay for average, but certain income

In other words:

Risk-averse people prefer a certain income over an uncertain income with the same average.

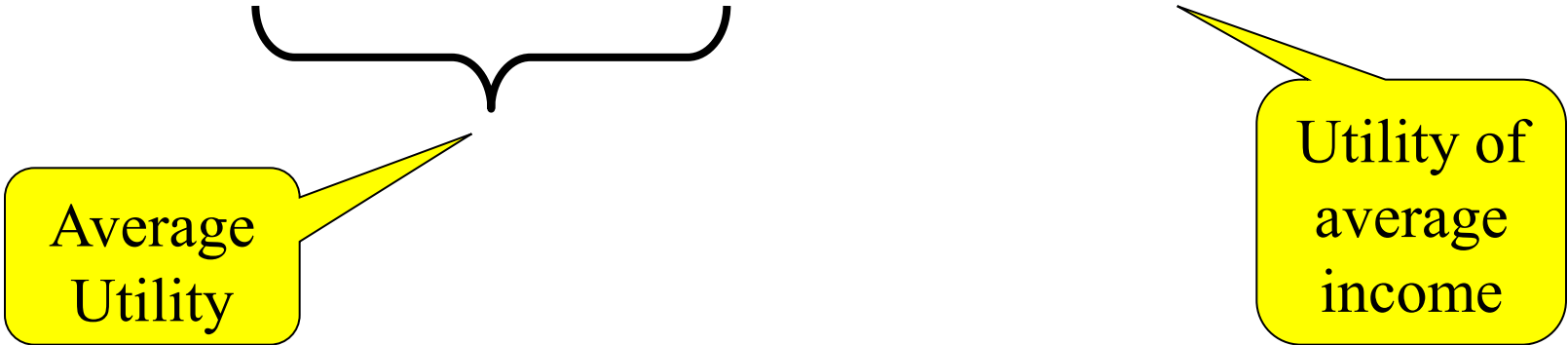
Application: insurance

Mathematically: risk aversion implies concave utility from income

$$U(Y); U'(Y) > 0; U''(Y) < 0$$

Suppose income Y_s , loss L with probability π

Then: $(1 - \pi)U(Y_s) + \pi U(Y_s - L) < U(Y_s - \pi L)$

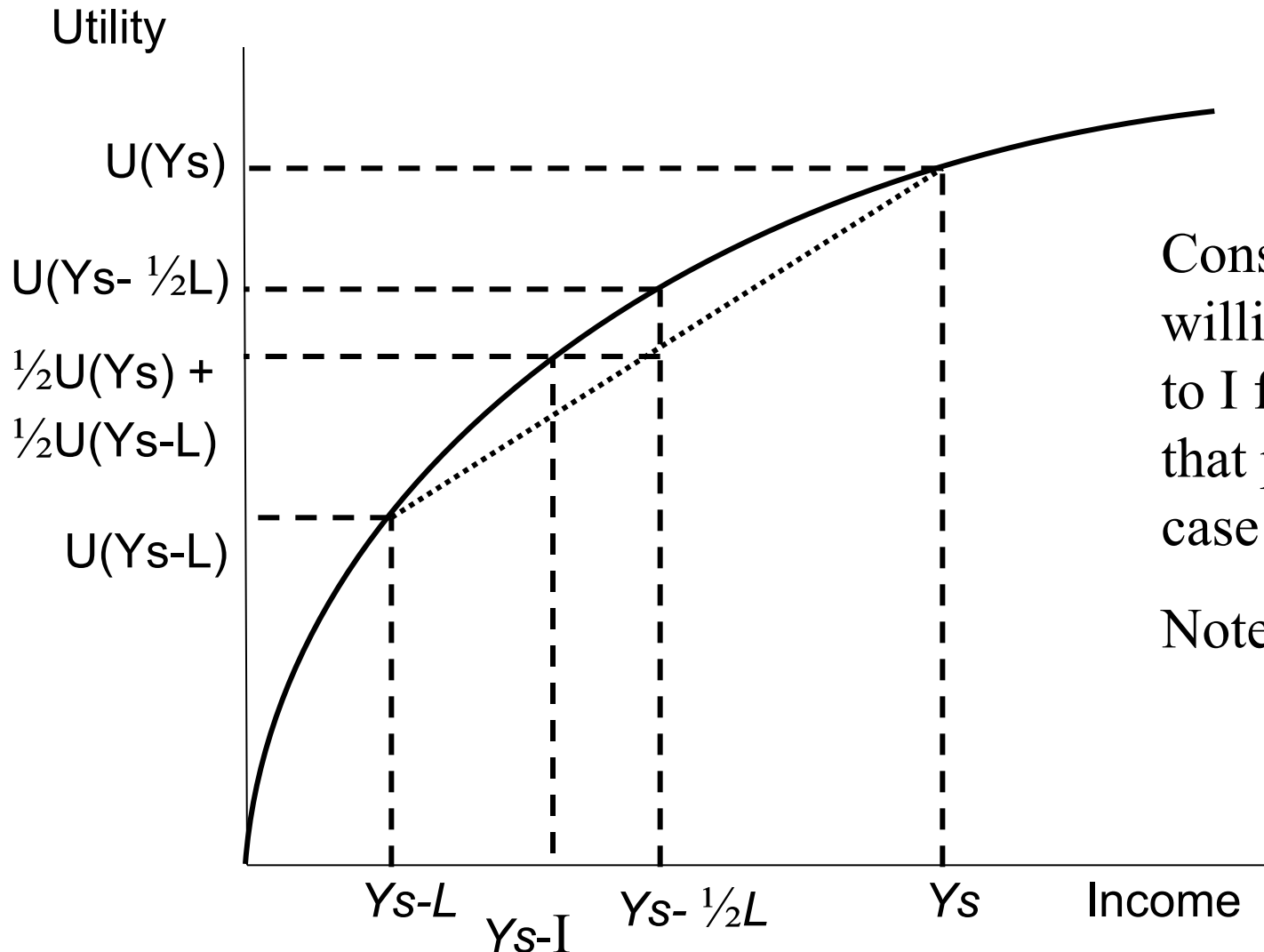


Average
Utility

Utility of
average
income

Application: insurance

Graphical proof, with $\pi = \frac{1}{2}$ (See also Rosen, app ch. 8)



Consumer is willing to pay up to I for insurance that pays out L in case of a loss.

Note: $I > \frac{1}{2} L$

Application: insurance

Insurance works by pooling many uncorrelated risks

→ If everyone pays expected (= average) loss in advance, then by the law of large numbers, this should be about enough to compensate those who actually ‘lose’

(an aside: risks must be uncorrelated, otherwise actual payout either very small or very large

→ No private insurance for natural disasters)

Application: insurance

Two main ‘problems’ in case of insurance:

1. Probability and size of loss depend on choices / behavior.
People may change behavior after receiving insurance,
which makes expected loss larger
→ Moral hazard
2. Individuals may know more about characteristics that
affect expected loss than insurance company, and insurance
is more valuable for individuals with large expected loss
→ adverse selection

Application: insurance

Problem: Probability and/or level of loss may be affected by individuals' characteristics or behaviour

→ **Adverse selection and moral hazard.**

Result: Market failure

- Markets do provide some insurance, but are not efficient
 - Before 'Obamacare', 40 million people without health insurance in the US:
- Markets do not provide insurance at all
 - Private poverty insurance does not exist

Application: insurance

Examples:

1. Car insurance
 - MH: drinking; speeding; etc.
 - AS: insurance most beneficial for bad drivers
2. Health insurance
 - MH: unhealthy lifestyle; risky hobbies
 - AS: full coverage most interesting for those who need costly treatments
3. (Hypothetical) Poverty insurance
 - MH: Insured have less incentive to work or train
 - AS: People with low productivity benefit most

Insurance: example moral hazard

- Viola has income $Y = 625$, probability $\frac{1}{2}$ on health care cost $H = 225$
 - Utility equals $\sqrt{C}$, where $C = \text{consumption} = Y - H$
 - If she exercises, probability reduces to $\frac{2}{5}$.
But exercising is tiring, and reduces utility by 0.4
1. Will she exercise?

Yes, as:

$$EU^{ne} = \frac{1}{2}\sqrt{625} + \frac{1}{2}\sqrt{625 - 225} = 22.5$$

$$EU^e = \frac{3}{5}\sqrt{625} + \frac{2}{5}\sqrt{625 - 225} - 0.4 = 22.6$$

Insurance: example moral hazard

Now: Suppose insurance company expects that Viola exercises, and offers insurance at expected cost ('actuarially fair')

→ premium $I = 2/5 * 225 = 90$; payout 225 when 'loss'

2. Will Viola accept insurance?

→ Yes: $U_I^e = \sqrt{625 - 90} - 0.4 = 22.73 > 22.6 = EU^e$

However, this is assuming Viola exercises!

3. Will Viola exercise?

→ **No!** $U_I^{ne} = \sqrt{625 - 90} = 23.13$

→ moral hazard!

Insurance: example moral hazard

So, premium cannot be 90 in equilibrium

→ insurance company makes a loss on average

Suppose insurance company correctly anticipates that Viola will not exercise when she has insurance:

→ premium $I = \frac{1}{2} * 225 = 112.5$; payout 225 when 'loss'

4. Will Viola accept insurance? → Yes:

$$U_I^{ne} = \sqrt{625 - 112.5} = 22.64 > 22.6 = EU^e$$

(Note: if exercise cost < 0.36 , Viola prefers no insurance!)

Observe that Viola would be better off if she could (credibly) commit to exercise!

→ Moral hazard reduces efficiency

Insurance: example moral hazard

Do-it-yourself:

- At which level of deductible would Viola choose to exercise if she has insurance?
(note: if the deductible is D , Viola gets payout $225 - D$ if she gets sick)
- Does an insurance with deductible yield higher utility?

Insurance: example adverse selection

Now, suppose Viola has twin sister Caroline

Same preferences, but her probability of incurring cost
 $H = 225$ is only 0.1, independent of exercising

If insurer can distinguish between the twins, premium for
Caroline equals $0.1 * 225 = 22.5$

Caroline will take up the insurance, as

$$U_I^{ne} = \sqrt{625 - 22.5} = 24.55 > 24.5 = 0.9 * \sqrt{625} + 0.1 * \sqrt{400}$$

Insurance: example adverse selection

Now suppose insurer cannot distinguish between the twins

→ Same premium

If both insured, exp. payout per person equals $0.3 \cdot 225 = 67.5$

However, Caroline will not take up the insurance, as

$$U_I^{ne} = \sqrt{625 - 67.5} = 23.61 < 24.5 = 0.9 \cdot \sqrt{625} + 0.1 \cdot \sqrt{400} = U_{NI}^e$$

→ Adverse selection: As the insurer cannot offer different ‘prices’ to the different sisters, the least costly sister is not willing to insure. Only the most costly sister self-selects into this transaction.

Risk-selection

On average, groups of people differ in how much they claim from their insurance company

Competition between insurance companies forces them to charge higher premium to relatively high-risk groups.

- Young people and people in densely populated area's pay higher car insurance premium
- Older people pay more for health insurance

Clearly: only possible with observable characteristics

On efficiency grounds, risk-selection makes sense:

- It reduces the adverse selection problem.

Risk-selection

However, you get into distributional and ethical issues:

- Should overweight people pay more for health insurance?
- Should people be tested on HIV before health insurance premium is being determined?

Dutch research (RIVM, 2007):

Poor people, divorced people, and people from ethnic minorities have significantly higher health care cost.

Compared to average: 21%, 48%, and 15% more, resp.

So, should these groups pay more for health insurance?

→ trade-off between efficiency and distribution

Risk-selection

Current Dutch solution: ‘Basisverzekering’

Idea:

- Minimum set of insured treatments
- Every citizen is required to have at least this insurance (reduces adverse selection problem; paternalism; ‘health insurance mandate’ in ‘Obamacare’ big issue in US)
- (Private) insurers have to accept everyone who wants this minimum insurance, one rate for all clients
→ no risk selection for the basisverzekering
- Compensation for insurers with large proportion of high-risk clients (‘risico-vereveningsfonds’)

Voorbeeld risicoverevening (in euro per jaar)

Verzekerde Criteria	Vrouw, 40, arbeids- ongeschikt, lage SES, ongunstige regio, diabetes-1	Man, 38, werknemer, hoge SES, gunstige regio, geen chronische ziekte
Leeftijd/geslacht	1182	946
Aard van het inkomen	965	-/- 63
Sociaal economische status	51	-/- 67
Regio	60	-/- 98
Farmaceutische kostengroep	3117	-/- 213
Diagnose kostengroep	-/- 98	-/- 98
Totale kosten	5277	407
Rekenpremie	-/- 970	-/- 970
Uitkering vereveningsfonds	4307	-/- 563

Concluding remarks

Asymmetric info may strongly reduce market efficiency

- Moral hazard: individuals take non-verifiable actions that are individually rational but socially inefficient
- Adverse selection: uninformed parties interact with the ‘worst’ fraction of the parties with better info on characteristics of transaction.

Potential private solutions to this market failure:

- Signaling and screening. But: costly
- Limiting possible choices / actions.

Potential public solutions:

- Regulation (information disclosure; mandatory insurance)
- Public production (NHS in the UK; ‘poverty insurance’)

Coming up

Next week: Redistribution

Conceptually:

Why would we want redistribution?

In practice:

Why is there so much public involvement in redistribution?

What are the efficiency effects of redistribution?